Grade 8 | Name_____________________
Unit 3 | Date_____________________
Lesson 3 Practice | Period_____________________

1. What is the value of 10^{-16}, written as a fraction?

2. Complete the statements.

The expression $\frac{10^{-17}}{10^{-5}}$ can be written as ______. Another way to write the expression $\frac{10^{-17}}{10^{-5}}$ using positive exponents only is _________.

3. Complete the table. Write each mathematical statement in an equivalent form by applying one or more laws of exponents. Then, write the equivalent decimal value.

Mathematical Statement	**Value Written as a Power of 10**	**Decimal Equivalent**
$(10^4) - 4 \times 10^{25}$		
$(\frac{1}{10^{-5}})^3$		
$\frac{10^{-14}}{10^{-9}} \cdot 10^6$		

4. Numbers like one-trillionth can be written using powers of 10. Write one-trillionth, or $\frac{1}{1{,}000{,}000{,}000{,}000}$ as a power of 10.

5. One octillionth can be written as $0.000000000000000000000000001$. Write one octillionth as a power of 10.

Using a calculator, find the value of each mathematical statement. Then, if possible, rewrite the value as a power of 10 or write 'not possible'.

Mathematical Statement	**Value Found Using the Calculator**	**Value Written as a Power of 10**
6. $(10^{-2})^5$		
7. $10^{-4} - 10^{-1}$		
8. $\frac{1}{10^{-9}}$		
9. $\frac{10^{-4}}{10^{-6}}$		
10. $10^{-6} + 10^{-2}$		